

QuickVet

AI-Powered Veterinary Healthcare Platform

Technical Documentation

Hackathon Project Documentation

Powered By

Temporal AI || Aikido || Kiro AI

Executive Summary

QuickVet is an AI-powered veterinary healthcare platform designed to make pet healthcare more accessible, reliable, and efficient. The platform connects pet owners with verified veterinary professionals, enabling emergency assistance, intelligent vaccination scheduling, appointment booking, clinic discovery, and a collaborative pet care community through a single unified application.

Unlike traditional veterinary platforms that primarily focus on appointment booking, QuickVet integrates artificial intelligence, workflow automation, cloud infrastructure, and secure software engineering practices to improve the complete healthcare journey for pets. By leveraging intelligent automation, the platform reduces manual effort for veterinarians while ensuring pet owners never miss critical healthcare milestones such as vaccinations and follow-up appointments.

To achieve this, QuickVet combines modern web technologies with enterprise-grade tools including Temporal for durable workflow orchestration, Aikido Security for continuous vulnerability monitoring and secure development practices, and Kiro AI to accelerate software development through AI-assisted engineering.

The result is a scalable, secure, and intelligent veterinary platform that demonstrates how modern cloud-native technologies can improve preventive pet healthcare and emergency response.

1. Project Overview

Introduction

QuickVet is a full-stack AI-powered veterinary healthcare platform developed to simplify the interaction between pet owners and veterinary professionals. The platform addresses common challenges faced by pet parents, including locating trusted veterinary clinics, managing vaccination schedules, booking appointments, and receiving timely healthcare reminders.

The application provides an intuitive user experience that enables pet owners to create digital pet profiles, manage appointments, access verified veterinary clinics, and participate in a community-driven ecosystem where pet care knowledge can be shared.

Healthcare providers benefit from streamlined appointment management, centralized patient records, and automated scheduling workflows, allowing them to focus more on providing quality care rather than administrative tasks.

Vision

To build an intelligent, secure, and scalable veterinary healthcare ecosystem that enhances preventive care, reduces emergency response delays, and improves communication between pet owners and veterinary professionals.

Objectives

The primary objectives of QuickVet are:

- Simplify veterinary appointment booking
- Digitize pet healthcare records
- Automate vaccination reminders
- Improve emergency response coordination
- Connect users with verified veterinary clinics
- Build an engaging pet care community
- Demonstrate enterprise-grade workflow orchestration using Temporal
- Showcase secure software development using Aikido
- Accelerate development using AI-assisted engineering with Kiro

2. Problem Statement

Pet healthcare remains fragmented across multiple independent services, creating unnecessary complexity for pet owners and veterinary professionals.

Several challenges motivated the development of QuickVet:

Emergency Accessibility

During medical emergencies, pet owners often struggle to quickly identify nearby veterinary clinics capable of providing immediate treatment. Delays in locating appropriate care can significantly affect patient outcomes.

Missed Vaccinations

Vaccination schedules vary depending on species, breed, age, and medical history. Many pet owners rely on manual reminders or memory, increasing the likelihood of missed vaccinations and preventive treatments.

Appointment Management

Traditional appointment booking methods frequently involve phone calls, manual scheduling, and limited visibility into veterinarian availability, resulting in scheduling inefficiencies.

Lack of Centralized Records

Pet healthcare information—including appointments, vaccination history, prescriptions, and medical records—is often scattered across different clinics, making long-term healthcare management difficult.

Limited Community Support

Pet owners frequently seek advice regarding nutrition, behavior, grooming, and healthcare. Existing platforms rarely integrate professional veterinary guidance with community discussions.

Security Concerns

Healthcare applications process sensitive personal information and medical records. Without continuous security monitoring and secure development practices, applications remain vulnerable to credential exposure, dependency vulnerabilities, and common software attacks.

3. Solution

QuickVet addresses these challenges through an integrated AI-powered veterinary healthcare platform.

The platform centralizes multiple veterinary services into a single application, allowing users to manage every aspect of their pet's healthcare journey.

Key capabilities include:

- **Intelligent appointment booking**
- **Vaccination scheduling**
- **Automated reminder workflows**
- **Verified veterinary discovery**
- **Secure user authentication**
- **Community interaction**
- **Cloud-based image management**
- **AI-assisted workflow automation**

QuickVet leverages enterprise-grade workflow orchestration using Temporal, ensuring long-running healthcare workflows remain reliable even during server failures.

Security is strengthened through Aikido, which continuously scans dependencies, source code, secrets, and deployment environments to detect vulnerabilities before production deployment.

Development productivity is significantly enhanced through Kiro AI, enabling rapid component generation, intelligent debugging, code refactoring, and documentation assistance throughout the project lifecycle.

4. Key Features

Emergency Veterinary Assistance

QuickVet enables pet owners to rapidly request emergency veterinary care.

Features

- **Emergency assistance requests**
- **Nearby veterinary discovery**
- **Live appointment status**
- **Emergency contact management**
- **Fast response workflow**

Smart Vaccination Management

Vaccination scheduling is one of the platform's core features.

Users can:

- **Schedule vaccinations**
- **Receive automated reminders**
- **View upcoming vaccinations**
- **Track completed vaccinations**
- **Maintain vaccination history**

Temporal workflows automate reminder delivery without relying on traditional cron jobs, providing reliable long-term scheduling.

Appointment Booking

QuickVet simplifies appointment scheduling by allowing users to:

- Browse available veterinarians
- Select appointment slots
- Book appointments online
- Receive confirmation notifications
- Manage appointment history

Veterinary Discovery

Users can search verified veterinary clinics based on:

- Location
- Availability
- Specialization
- Ratings
- Reviews

This reduces the effort required to locate trusted veterinary professionals.

Community Platform

QuickVet includes a social platform where pet owners can:

- Share experiences
- Ask healthcare questions
- Provide recommendations
- Discuss pet nutrition
- Exchange grooming tips
- Connect with other pet owners

Secure Authentication

Authentication is implemented using:

- Google OAuth
- JWT Authentication
- Secure password hashing
- Protected API routes
- Role-based access control

Cloud Image Management

Pet images and medical documents are securely stored using Cloudinary.

Benefits include:

- Fast CDN delivery
- Image optimization
- Secure storage
- Reduced backend storage requirements

Automated Notifications

The platform automatically sends:

- Appointment confirmations
- Vaccination reminders
- Booking updates
- Account notifications
- Email verification messages

These notifications are orchestrated using Temporal workflows, ensuring reliable delivery even in the event of temporary service interruptions.

Dashboard

Users have access to personalized dashboards displaying:

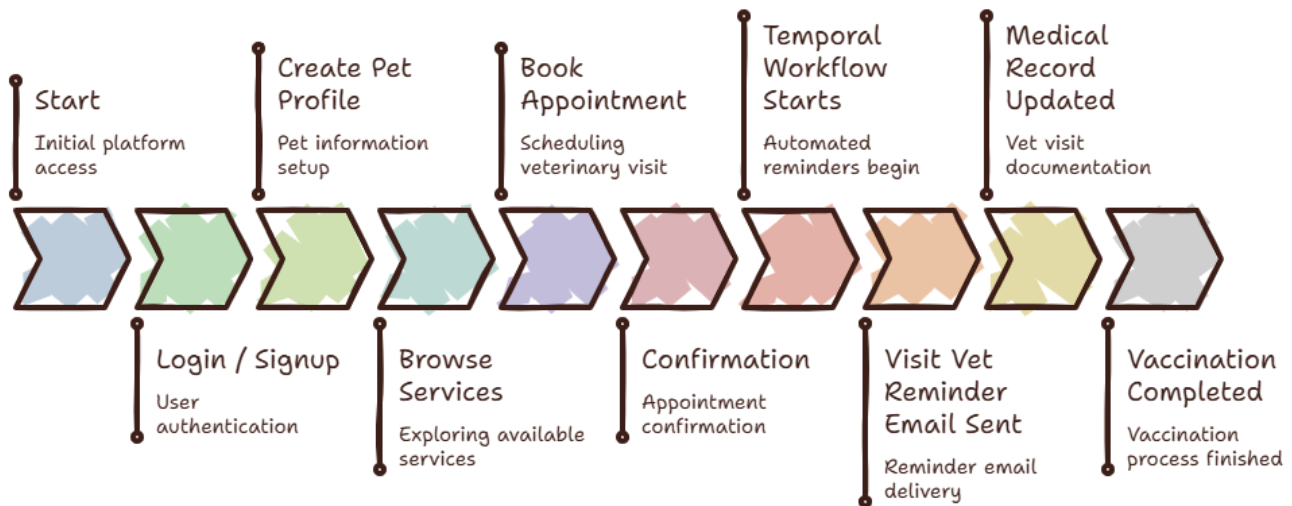
- Upcoming appointments
- Vaccination schedules
- Medical history
- Community activity
- Pet profiles

Administrators and veterinarians can manage appointments, monitor bookings, and oversee platform activity through dedicated management interfaces.

5. User Journey

Pet Owner Workflow

Pet Healthcare Journey: From Start to Completion

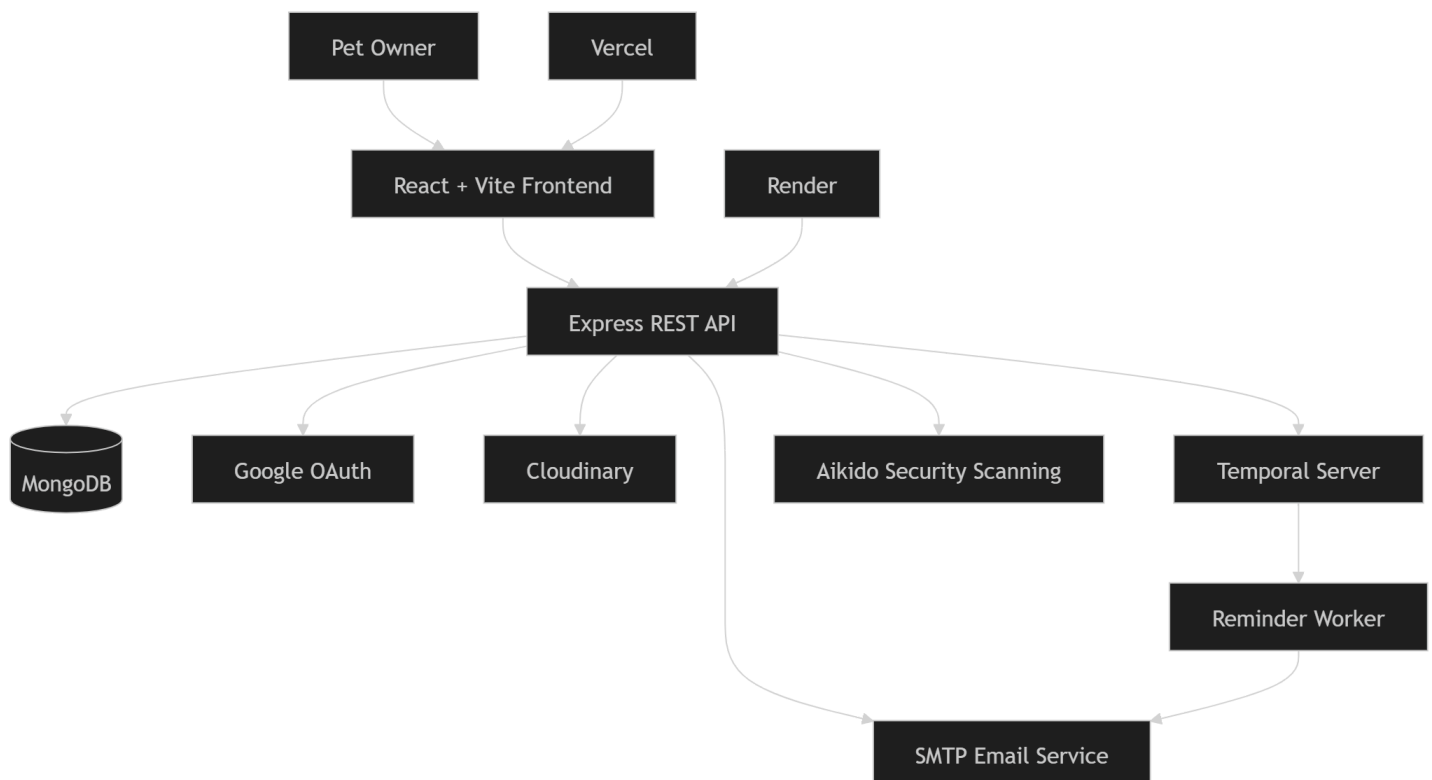


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6. Overall System Architecture

QuickVet follows a modern cloud-native architecture consisting of a React frontend, an Express.js backend, MongoDB for persistent storage, Temporal for workflow orchestration, and Cloudinary for media storage.

Architecture Overview



Architecture Components

Frontend Layer

Technology

- React
- Vite
- TypeScript
- Tailwind CSS
- Framer Motion

Responsibilities

- User Interface
- Authentication
- Appointment Booking
- Vaccination Scheduling
- Community Features
- Dashboard
- Emergency Requests

Backend Layer

Technology

- Node.js
- Express.js

Responsibilities include:

- Authentication
- REST APIs
- Appointment Management
- Vaccination Scheduling
- Community APIs
- Pet Management
- Admin APIs
- Notification Triggering

Database Layer

MongoDB stores all persistent application data.

Collections include:

Veterinary Clinic Management System Components



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Workflow Layer

Temporal handles long-running healthcare workflows.

Examples include:

- Vaccination reminders
- Appointment reminders
- Retry mechanisms
- Delayed notifications
- Durable timers

Storage Layer

Cloudinary stores:

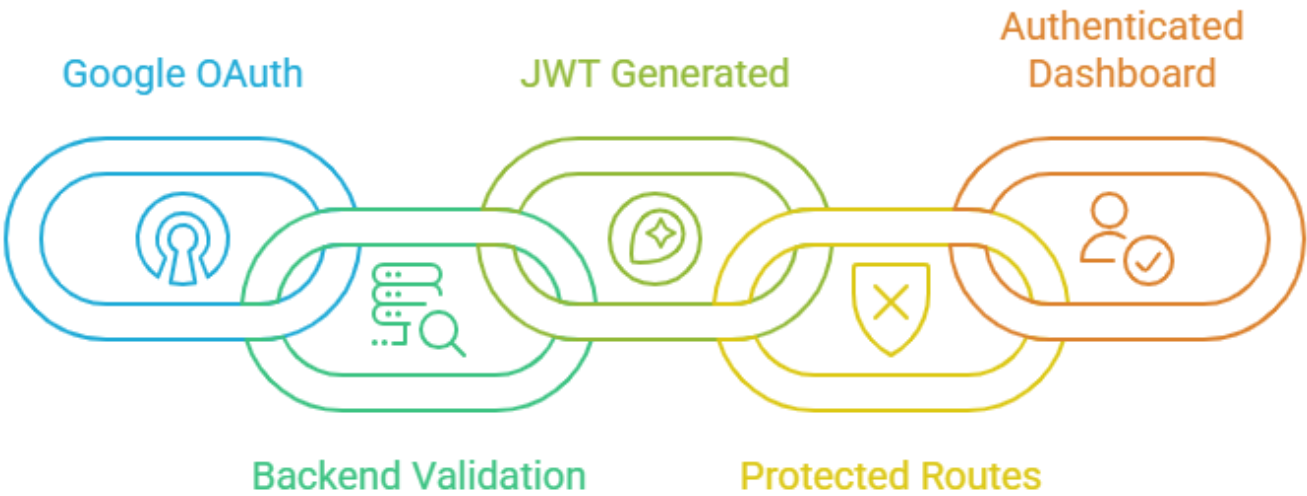
- Pet images
- Medical reports
- User profile pictures
- Community uploads

Authentication Layer

Authentication uses Google OAuth together with JWT.

Flow:

Authentication Flow



Layer	Technology
Frontend	React
Build Tool	Vite
Styling	Tailwind CSS
Language	TypeScript
Animations	Framer Motion
Backend	Node.js
Framework	Express
Database	Supabase
Authentication	Google OAuth + JWT
Image Storage	Cloudinary
Workflow Engine	Temporal
Email Service	SMTP
Deployment	Vercel + Render
Security	Aikido
Version Control	GitHub
AI Development	Kiro AI

8. Database Design

QuickVet uses Supabase because of its flexibility for handling healthcare-related data structures.

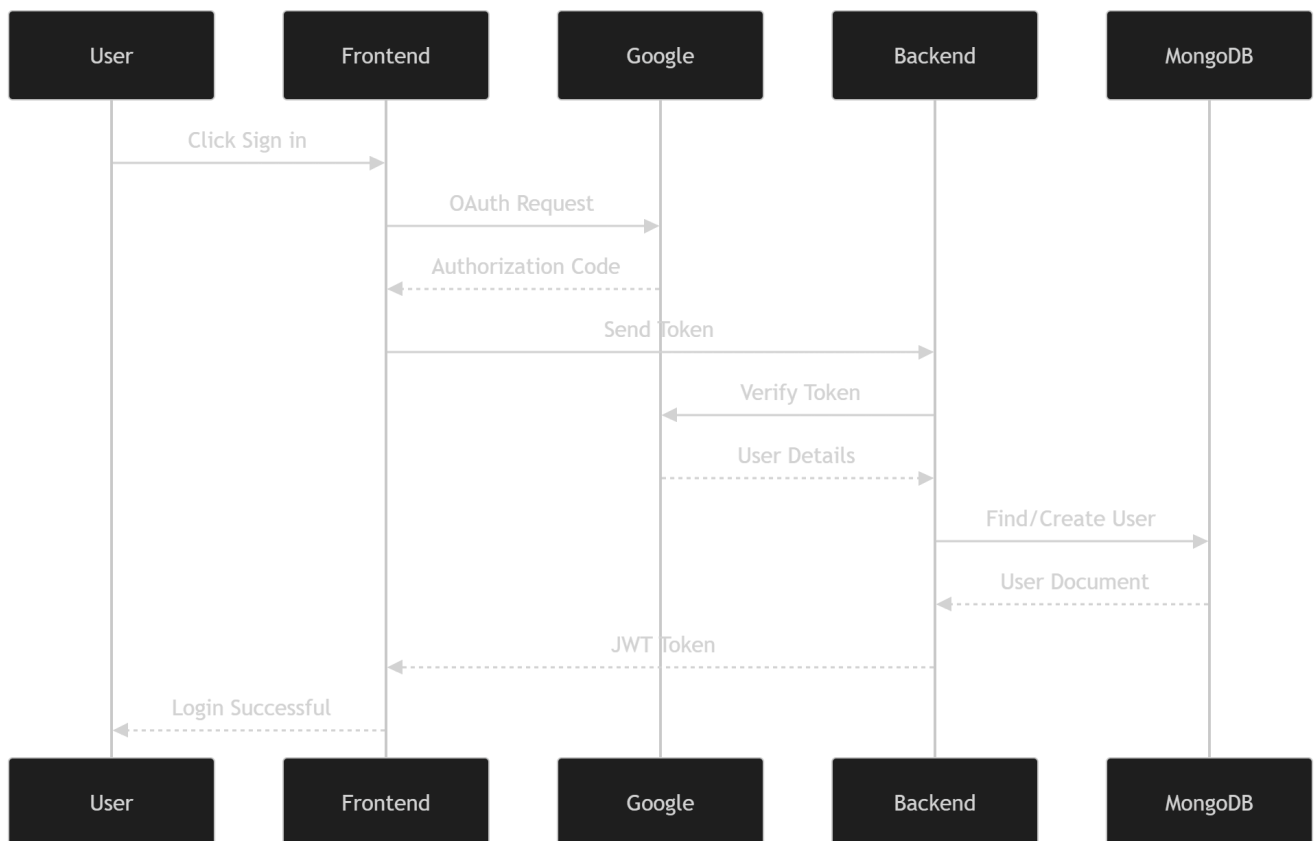
Entity Relationship Overview

Comprehensive Pet Care Overview



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9. Authentication Flow



10. Artificial Intelligence Workflow

AI-Powered Healthcare Automation

QuickVet integrates artificial intelligence and workflow orchestration to automate routine healthcare tasks while keeping veterinarians in control of medical decisions. The AI layer assists with validation, scheduling, reminders, and workflow initiation, allowing pet owners to receive timely notifications and reducing administrative overhead for veterinary clinics.

Rather than replacing veterinarians, AI acts as an intelligent assistant that helps streamline preventive healthcare and appointment management.

AI Workflow Overview

Pet Owner Books
Vaccination



Frontend Validation



Backend API



Validate Pet Information



AI Recommendation Engine



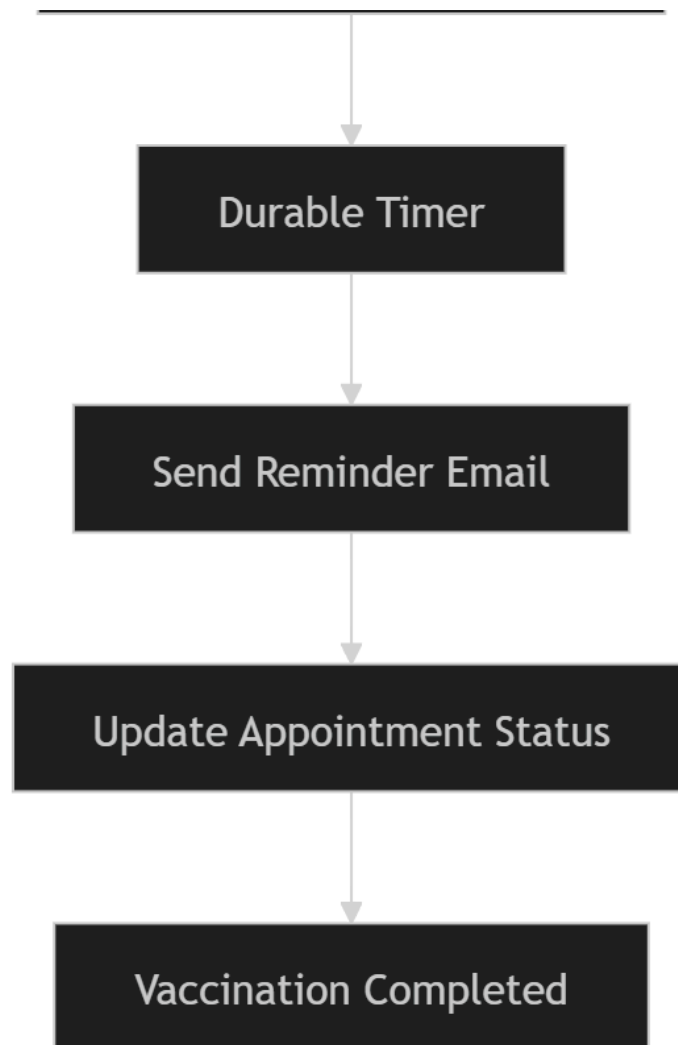
Determine Reminder
Schedule



Store Appointment in
MongoDB



Start Temporal Workflow



AI Responsibilities

The AI layer assists with:

- Appointment validation
- Vaccination schedule recommendations
- Reminder timing
- Automated workflow initiation
- Intelligent notification management
- Data consistency checks
- Administrative task automation

AI Workflow Benefits

- Reduces manual scheduling
- Prevents missed vaccinations
- Improves appointment accuracy
- Minimizes administrative workload
- Enables proactive pet healthcare

11. Temporal AI Integration

Why Temporal?

Many applications use cron jobs or background schedulers to send reminders. While suitable for simple tasks, these approaches become unreliable when workflows span days or weeks.

Pet healthcare often requires reminders long after an appointment is created. For example:

- Tomorrow
- Next week
- One month later
- Annual booster vaccinations

Traditional schedulers struggle with long-running processes because they depend on server uptime. If the server crashes or restarts, scheduled jobs can be lost.

Temporal solves this problem by providing durable workflow execution, ensuring that healthcare reminders remain reliable regardless of server failures.

Why We Chose Temporal

QuickVet uses Temporal because it provides:

- Durable execution
- Long-running workflows
- Automatic retries
- Built-in timers
- State persistence
- Fault tolerance
- Workflow observability

This makes it an ideal solution for healthcare applications where reliable scheduling is critical

Detailed Workflow Explanation

Step 1 — Appointment Booking

The pet owner selects a veterinarian, date, and vaccination type.

The backend validates:

- Pet details
- Appointment availability

- Vaccination schedule
- Required information

Only after successful validation is the appointment stored.

Step 2 — Workflow Creation

Immediately after booking:

- Appointment stored in MongoDB
- Workflow ID generated
- Temporal workflow started
- Timer initialized

Unlike cron jobs, no continuously running scheduler is required.

Step 3 — Durable Waiting

Temporal now "sleeps" until the reminder date.

Example:

Appointment:

15 August

Reminder:

13 August

Temporal stores:

- Current state
- Timer
- Workflow progress

Even if the backend server restarts several times, the workflow resumes automatically.

Step 4 — Reminder Execution

When the reminder time arrives:

Worker wakes up

↓

Fetches appointment

↓

Generates reminder



Sends email



Updates database



Continues workflow

Step 5 — Appointment Completion

After the scheduled appointment:

- **Appointment status updated**
- **Vaccination history recorded**
- **Workflow marked completed**
- **Resources released**

Durable Execution

One of Temporal's most valuable features is durable execution.

Suppose the workflow is waiting for 20 days.

Day 10:

Server crashes.

Traditional scheduler:

 **Reminder lost**

Temporal:

 **Workflow state restored**

 **Timer resumes**

 **Reminder delivered successfully**

This eliminates one of the biggest reliability challenges in long-running healthcare systems.

Benefits to QuickVet

By integrating Temporal, QuickVet achieves:

- Reliable vaccination reminders
- Durable appointment workflows
- Automatic recovery from failures
- Built-in retry mechanisms
- Scalable background processing
- Improved user trust
- Reduced operational complexity

Temporal enables QuickVet to deliver enterprise-grade workflow reliability, ensuring that critical healthcare reminders are never missed—even during server failures or temporary service outages.

12. Aikido Security Integration

Why Security Matters

QuickVet manages sensitive information that must be protected throughout the application's lifecycle.

The platform stores and processes:

- User accounts
- Pet profiles
- Veterinary appointments
- Vaccination records
- Clinic information
- Authentication credentials
- Cloud storage configurations
- Email service credentials

Protecting this data is essential for maintaining user trust and ensuring the platform is secure against common software vulnerabilities.

To strengthen the security posture of the project, QuickVet integrates Aikido Security, providing continuous security analysis during development and deployment.

What is Aikido?

Aikido is an Application Security Platform that helps developers identify vulnerabilities before they reach production.

Unlike traditional security tools that require dedicated security teams, Aikido integrates directly into the development workflow and continuously analyzes the application for potential risks.

QuickVet leverages Aikido to ensure that the project follows secure development practices from the beginning of the software lifecycle.

Security Features Used

1. Dependency Scanning

Modern applications rely heavily on third-party packages.

QuickVet uses libraries for:

- React
- Express
- PostgreSQL
- JWT
- Google OAuth
- Cloudinary
- SMTP
- Temporal

A vulnerable dependency could expose the entire application.

Aikido continuously scans project dependencies and identifies:

- Known CVEs
- Outdated packages
- High-risk libraries
- Supply-chain vulnerabilities

Benefits

- Prevents vulnerable packages from reaching production
- Encourages timely dependency updates
- Reduces software supply-chain risks

3. Static Application Security Testing (SAST)

Aikido performs static code analysis to detect insecure coding practices before the application is deployed.

Typical vulnerabilities that can be detected include:

- SQL/NoSQL Injection risks
- Cross-Site Scripting (XSS)
- Command Injection
- Unsafe input handling
- Weak authentication implementations

- Insecure API endpoints

This allows developers to resolve issues early in the development cycle.

4. Container & Deployment Security

Although QuickVet is deployed using cloud platforms, deployment artifacts and runtime environments should still be evaluated for security.

Aikido helps identify:

- Vulnerable base images
- Insecure configurations
- Outdated runtime components
- Known operating system vulnerabilities

This contributes to a more secure production environment.

5. Continuous Security Monitoring

Security is not treated as a one-time activity.

Each update to the project can be automatically analyzed to identify newly introduced vulnerabilities.

This continuous monitoring approach supports long-term maintainability and reduces the likelihood of unnoticed security regressions.

Security Best Practices Implemented

QuickVet follows several secure development practices in addition to automated scanning.

Authentication

- Google OAuth integration
- JWT-based session management
- Secure password hashing using bcrypt
- Protected API routes

API Security

- Input validation
- Error handling
- Authentication middleware
- Route authorization
- Request sanitization

Security Benefits for QuickVet

By integrating Aikido into the development workflow, QuickVet gains:

- Early vulnerability detection
- Continuous dependency monitoring
- Secret leak prevention
- Improved deployment confidence
- Better software quality
- Reduced security risks
- More secure development lifecycle

Future Security Enhancements

Potential future improvements include:

- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Audit logging
- Security Information and Event Management (SIEM)
- Rate limiting for APIs
- Advanced anomaly detection
- Compliance with veterinary healthcare regulations

Conclusion

QuickVet demonstrates how modern technologies can be combined to create a secure, scalable, and intelligent veterinary healthcare platform. By integrating AI-assisted development, durable workflow orchestration, and secure software engineering practices, the platform streamlines appointment management, automates vaccination reminders, and improves access to veterinary care for pet owners.

The project showcases the practical use of Temporal for reliable long-running workflows, Aikido Security for continuous vulnerability detection and secure development practices, and Kiro AI for accelerating development through AI-assisted coding, debugging, and documentation. Together, these technologies enabled the team to build a production-ready solution with improved reliability, maintainability, and security.

Although QuickVet was developed as a hackathon project, its modular architecture provides a strong foundation for future enhancements such as AI-powered symptom analysis, telemedicine, digital pet health records, wearable device integration, and intelligent healthcare analytics. With further development, QuickVet has the potential to become a comprehensive digital healthcare ecosystem that improves the quality and accessibility of veterinary services

Team

Member	Role
Harshit Garg	Full Stack Developer
Abhishek Gupta	Full Stack Developer

Thank You

QuickVet is more than just a veterinary appointment platform—it's a step toward smarter, more connected, and technology-driven pet healthcare.

Thank you for reviewing our project. We hope QuickVet demonstrates the potential of combining AI, workflow automation, and secure engineering to improve the future of veterinary care.